

TWO-WATT TBA820M AF AMPLIFIER

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Here is a simple audio-frequency amplifier project based on TBA820M IC. This amplifier can deliver two-watt output power with 8-ohm speaker

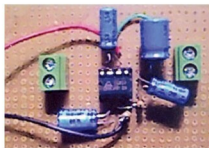


Fig. 1: Author's prototype

PARTS LIST

Semiconductors:	
IC1	- TBA820M audio amplifier
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 5.6-kilo-ohm
R2	- 47-ohm
VR1	- 10-kilo-ohm potmeter
Capacitors:	
C1	- 0.1µF ceramic disk
C2	- 0.001µF ceramic disk
C3	- 100µF, 25V electrolytic
C4	- 1µF, 25V electrolytic
C5, C7	- 220µF, 25V electrolytic
C6	- 47µF, 25V electrolytic
Miscellaneous:	
LS1	- 8-ohm, 2W speaker
CON1	- 2-pin connector
CON2	- 2-pin terminal connector
S1	- On/off switch
	- 12V DC power supply/12V battery

using a 12V power supply. A 3-16V battery or an AC/DC adaptor can be used to power this circuit. Author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the two-watt TBA820M AF amplifier is shown in Fig. 2. An 8-ohm, 2W loudspeaker is connected to output pin 5 of IC1 through C5. Pin 8 (ripple rejection) is connected to +Vcc through C6. Also, C7 is connected to positive and ground for noise filtering.

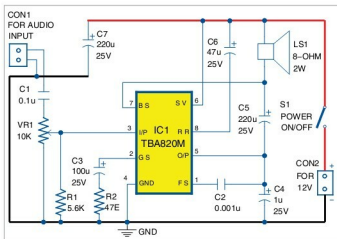


Fig. 2: Circuit diagram of the two-watt TBA820M AF amplifier

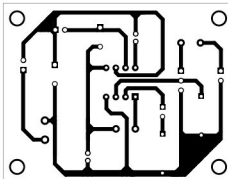


Fig. 3: Actual-size PCB pattern of the two-watt TBA820M AF amplifier

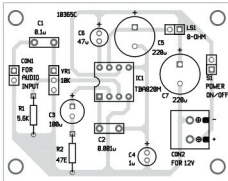


Fig. 4: Component layout of the PCB

Construction and testing

An actual-size, single-side PCB pattern for the amplifier circuit is

shown in Fig. 3 and its component layout in Fig. 4. A general-purpose PCB or veroboard can also be used to build this circuit.

It is better to use an IC socket for the IC. After assembling the circuit on the PCB, enclose

it in a suitable plastic case.

For input and output connections, you may use 2-pin connectors. This will make the project neat and clean for easy connections.

After assembling the circuit, calibrate the circuit. For calibration, connect a 12V battery. First, check correct polarity of the power supply and then switch on S1. Then, connect the speaker to output of the circuit. Set potmeter VR1 to its mid position.

Take a small screwdriver and gently touch input pin 3 of TBA820M. You will hear a humming sound from the speaker if the circuit is working. Connect audio source like MP3 player or mobile phone at CON1. Adjust volume control VR1 until you get a loud and clear sound. **EFY**



Raj K. Gorkhali is a regular contributor to EFY. His interests include designing electronics circuit